Tensor model server installation packages

```
!echo "deb http://storage.googleapis.com/tensorflow-serving-apt stable
tensorflow-model-server tensorflow-model-server-universal" | tee
/etc/apt/sources.list.d/tensorflow-serving.list && \
curl https://storage.googleapis.com/tensorflow-serving-apt/tensorflow-
serving.release.pub.gpg | apt-key add -
!apt update
```

```
deb http://storage.googleapis.com/tensorflow-serving-apt stable
tensorflow-model-server tensorflow-model-server-universal
  % Total    % Received % Xferd  Average Speed   Time    Time     Time
Current
                                 Dload  Upload   Total   Spent    Left
Speed
100  2943  100  2943    0     0  31989      0 --:--:-- --:--:--
--:--:-- 31989
OK
Get:1 http://storage.googleapis.com/tensorflow-serving-apt stable
InRelease [3,012 B]
Get:2 https://cloud.r-project.org/bin/linux/ubuntu bionic-cran40/
InRelease [3,626 B]
Ign:3
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/
x86_64  InRelease
Ign:4
https://developer.download.nvidia.com/compute/machine-learning/repos/
ubuntu1804/x86_64  InRelease
Hit:5 http://archive.ubuntu.com/ubuntu bionic InRelease
Get:6 http://security.ubuntu.com/ubuntu bionic-security InRelease
[88.7 kB]
Get:7
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/
x86_64  Release [696 B]
Get:8 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu bionic
InRelease [15.9 kB]
Hit:9
https://developer.download.nvidia.com/compute/machine-learning/repos/
ubuntu1804/x86_64  Release
Get:10
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/
x86_64  Release.gpg [836 B]
Get:11 http://storage.googleapis.com/tensorflow-serving-apt
stable/tensorflow-model-server amd64 Packages [341 B]
Get:12 http://archive.ubuntu.com/ubuntu bionic-updates InRelease [88.7
kB]
Get:13 http://storage.googleapis.com/tensorflow-serving-apt
stable/tensorflow-model-server-universal amd64 Packages [349 B]
Get:14 https://cloud.r-project.org/bin/linux/ubuntu bionic-cran40/
Packages [80.8 kB]
```

```
Hit:15 http://ppa.launchpad.net/cran/libgit2/ubuntu bionic InRelease
Get:16 http://archive.ubuntu.com/ubuntu bionic-backports InRelease
[74.6 kB]
Get:17 http://ppa.launchpad.net/deadsnakes/ppa/ubuntu bionic InRelease
[15.9 kB]
Get:19
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/
x86_64  Packages [950 kB]
Get:20 http://ppa.launchpad.net/graphics-drivers/ppa/ubuntu bionic
InRelease [21.3 kB]
Get:21 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu
bionic/main Sources [1,830 kB]
Get:22 http://security.ubuntu.com/ubuntu bionic-security/restricted
amd64 Packages [884 kB]
Get:23 http://archive.ubuntu.com/ubuntu bionic-updates/restricted
amd64 Packages [918 kB]
Get:24 http://security.ubuntu.com/ubuntu bionic-security/universe
amd64 Packages [1,486 kB]
Get:25 http://security.ubuntu.com/ubuntu bionic-security/main amd64
Packages [2,693 kB]
Get:26 http://archive.ubuntu.com/ubuntu bionic-updates/multiverse
amd64 Packages [29.8 kB]
Get:27 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64
Packages [3,132 kB]
Get:28 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu
bionic/main amd64 Packages [937 kB]
Get:29 http://archive.ubuntu.com/ubuntu bionic-updates/universe amd64
Packages [2,264 kB]
Get:30 http://archive.ubuntu.com/ubuntu bionic-backports/main amd64
Packages [12.2 kB]
Get:31 http://archive.ubuntu.com/ubuntu bionic-backports/universe
amd64 Packages [12.9 kB]
Get:32 http://ppa.launchpad.net/deadsnakes/ppa/ubuntu bionic/main
amd64 Packages [45.3 kB]
Get:33 http://ppa.launchpad.net/graphics-drivers/ppa/ubuntu
bionic/main amd64 Packages [44.3 kB]
Fetched 15.6 MB in 4s (3,922 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
82 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
!apt-get install tensorflow-model-server
```

```
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  tensorflow-model-server
0 upgraded, 1 newly installed, 0 to remove and 82 not upgraded.
Need to get 340 MB of archives.
```

```
After this operation, 0 B of additional disk space will be used.
Get:1 http://storage.googleapis.com/tensorflow-serving-apt
stable/tensorflow-model-server amd64 tensorflow-model-server all 2.8.0
[340 MB]
Fetched 340 MB in 6s (56.1 MB/s)
Selecting previously unselected package tensorflow-model-server.
(Reading database ... 156210 files and directories currently
installed.)
Preparing to unpack .../tensorflow-model-server_2.8.0_all.deb ...
Unpacking tensorflow-model-server (2.8.0) ...
Setting up tensorflow-model-server (2.8.0) ...
```

```
# import tensorflow-model-server
```

TFX installation

```
!pip install -U tfx
```

```
Collecting tfx
  Downloading tfx-1.7.0-py3-none-any.whl (2.5 MB)
-manylinux_2_12_x86_64.manylinux2010_x86_64.whl (19.2 MB)
ent already satisfied: numpy<2,>=1.16 in
/usr/local/lib/python3.7/dist-packages (from tfx) (1.21.5)
Collecting docker<5,>=4.1
  Downloading docker-4.4.4-py2.py3-none-any.whl (147 kB)
-manylinux_2_12_x86_64.manylinux2010_x86_64.whl (1.4 MB)
-manylinux2014_x86_64.whl (23.6 MB)
ent already satisfied: tensorflow-hub<0.13,>=0.9.0 in
/usr/local/lib/python3.7/dist-packages (from tfx) (0.12.0)
Requirement already satisfied: pyyaml<6,>=3.12 in
/usr/local/lib/python3.7/dist-packages (from tfx) (3.13)
Collecting kubernetes<13,>=10.0.1
  Downloading kubernetes-12.0.1-py2.py3-none-any.whl (1.7 MB)
<1.8.0,>=1.7.0
  Downloading tensorflow_transform-1.7.0-py3-none-any.whl (433 kB)
<2,>=1.6.2
  Downloading google_cloud_aiplatform-1.11.0-py2.py3-none-any.whl (1.8
MB)
ent already satisfied: absl-py<2.0.0,>=0.9 in
/usr/local/lib/python3.7/dist-packages (from tfx) (1.0.0)
Requirement already satisfied: portpicker<2,>=1.3.1 in
/usr/local/lib/python3.7/dist-packages (from tfx) (1.3.9)
Collecting tensorflow-model-analysis<0.39,>=0.38.0
  Downloading tensorflow_model_analysis-0.38.0-py3-none-any.whl (1.8
MB)
ent already satisfied: protobuf<4,>=3.13 in
/usr/local/lib/python3.7/dist-packages (from tfx) (3.17.3)
Requirement already satisfied: grpcio<2,>=1.28.1 in
/usr/local/lib/python3.7/dist-packages (from tfx) (1.44.0)
Collecting attrs<21,>=19.3.0
  Downloading attrs-20.3.0-py2.py3-none-any.whl (49 kB)
```

```
l-pipelines-sdk==1.7.0
  Downloading ml_pipelines_sdk-1.7.0-py3-none-any.whl (1.3 MB)
ent already satisfied: click<8,>=7 in /usr/local/lib/python3.7/dist-
packages (from tfx) (7.1.2)
Requirement already satisfied: tensorflow!=2.0.*,!=2.1.*,!=2.2.*,!
=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5 in
/usr/local/lib/python3.7/dist-packages (from tfx) (2.8.0)
Requirement already satisfied: jinja2<4,>=2.7.3 in
/usr/local/lib/python3.7/dist-packages (from tfx) (2.11.3)
Requirement already satisfied: google-api-python-client<2,>=1.8 in
/usr/local/lib/python3.7/dist-packages (from tfx) (1.12.11)
Collecting apache-beam[gcp]<3,>=2.36
  Downloading apache_beam-2.37.0-cp37-cp37m-manylinux2010_x86_64.whl
(10.1 MB)
l-metadata<1.8.0,>=1.7.0
  Downloading ml_metadata-1.7.0-cp37-cp37m-
manylinux_2_12_x86_64.manylinux2010_x86_64.whl (6.6 MB)
ent already satisfied: six in /usr/local/lib/python3.7/dist-packages
(from absl-py<2.0.0,>=0.9->tfx) (1.15.0)
Requirement already satisfied: pydot<2,>=1.2.0 in
/usr/local/lib/python3.7/dist-packages (from apache-
beam[gcp]<3,>=2.36->tfx) (1.3.0)
Requirement already satisfied: pytz>=2018.3 in
/usr/local/lib/python3.7/dist-packages (from apache-
beam[gcp]<3,>=2.36->tfx) (2018.9)
Collecting cloudpickle<3,>=2.0.0
  Downloading cloudpickle-2.0.0-py3-none-any.whl (25 kB)
Requirement already satisfied: typing-extensions>=3.7.0 in
/usr/local/lib/python3.7/dist-packages (from apache-
beam[gcp]<3,>=2.36->tfx) (3.10.0.2)
Collecting requests<3.0.0,>=2.24.0
  Downloading requests-2.27.1-py2.py3-none-any.whl (63 kB)
-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.3 MB)
ent already satisfied: crcmod<2.0,>=1.7 in
/usr/local/lib/python3.7/dist-packages (from apache-
beam[gcp]<3,>=2.36->tfx) (1.7)
Collecting proto-plus<2,>=1.7.1
  Downloading proto_plus-1.20.3-py3-none-any.whl (46 kB)
ent already satisfied: httplib2<0.20.0,>=0.8 in
/usr/local/lib/python3.7/dist-packages (from apache-
beam[gcp]<3,>=2.36->tfx) (0.17.4)
Requirement already satisfied: python-dateutil<3,>=2.8.0 in
/usr/local/lib/python3.7/dist-packages (from apache-
beam[gcp]<3,>=2.36->tfx) (2.8.2)
Requirement already satisfied: oauth2client<5,>=2.0.1 in
/usr/local/lib/python3.7/dist-packages (from apache-
beam[gcp]<3,>=2.36->tfx) (4.1.3)
Collecting orjson<4.0
  Downloading orjson-3.6.7-cp37-cp37m-manylinux_2_24_x86_64.whl (255
kB)
```

```
ongo<4.0.0,>=3.8.0
  Downloading pymongo-3.12.3-cp37-cp37m-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl (508 kB)
ent already satisfied: google-cloud-core<2,>=0.28.1 in
/usr/local/lib/python3.7/dist-packages (from apache-
beam[gcp]<3,>=2.36->tfx) (1.0.3)
Collecting google-cloud-dlp<4,>=3.0.0
  Downloading google_cloud_dlp-3.6.2-py2.py3-none-any.whl (113 kB)
ent already satisfied: cachetools<5,>=3.1.0 in
/usr/local/lib/python3.7/dist-packages (from apache-
beam[gcp]<3,>=2.36->tfx) (4.2.4)
Requirement already satisfied: google-auth<3,>=1.18.0 in
/usr/local/lib/python3.7/dist-packages (from apache-
beam[gcp]<3,>=2.36->tfx) (1.35.0)
Collecting google-cloud-bigquery-storage>=2.6.3
  Downloading google_cloud_bigquery_storage-2.13.0-py2.py3-none-
any.whl (179 kB)
ent already satisfied: google-cloud-datastore<2,>=1.8.0 in
/usr/local/lib/python3.7/dist-packages (from apache-
beam[gcp]<3,>=2.36->tfx) (1.8.0)
Collecting google-cloud-videointelligence<2,>=1.8.0
  Downloading google_cloud_videointelligence-1.16.1-py2.py3-none-
any.whl (183 kB)
mendations-ai<=0.2.0,>=0.1.0
  Downloading google_cloud_recommendations_ai-0.2.0-py2.py3-none-
any.whl (180 kB)
ent already satisfied: uritemplate<4dev,>=3.0.0 in
/usr/local/lib/python3.7/dist-packages (from google-api-python-
client<2,>=1.8->tfx) (3.0.1)
Requirement already satisfied: google-api-core<3dev,>=1.21.0 in
/usr/local/lib/python3.7/dist-packages (from google-api-python-
client<2,>=1.8->tfx) (1.26.3)
Requirement already satisfied: google-auth-httplib2>=0.0.3 in
/usr/local/lib/python3.7/dist-packages (from google-api-python-
client<2,>=1.8->tfx) (0.0.4)
Requirement already satisfied: googleapis-common-protos<2.0dev,>=1.6.0
in /usr/local/lib/python3.7/dist-packages (from google-api-
core<3dev,>=1.21.0->google-api-python-client<2,>=1.8->tfx) (1.56.0)
Requirement already satisfied: setuptools>=40.3.0 in
/usr/local/lib/python3.7/dist-packages (from google-api-
core<3dev,>=1.21.0->google-api-python-client<2,>=1.8->tfx) (57.4.0)
Collecting fasteners>=0.14
  Downloading fasteners-0.17.3-py3-none-any.whl (18 kB)
Requirement already satisfied: rsa<5,>=3.1.4 in
/usr/local/lib/python3.7/dist-packages (from google-auth<3,>=1.18.0-
>apache-beam[gcp]<3,>=2.36->tfx) (4.8)
Requirement already satisfied: pyasn1-modules>=0.2.1 in
/usr/local/lib/python3.7/dist-packages (from google-auth<3,>=1.18.0-
>apache-beam[gcp]<3,>=2.36->tfx) (0.2.8)
Collecting google-api-core[grpc]!=2.0.*,!=2.1.*,!=2.2.*,!
```

```
=2.3.0,<3.0.0dev,>=1.31.5
  Downloading google_api_core-2.7.1-py3-none-any.whl (114 kB)
able-media<3.0dev,>=0.6.0
  Downloading google_resumable_media-2.3.2-py2.py3-none-any.whl (76
kB)
-v1<0.13dev,>=0.12.3
  Downloading grpc-google-iam-v1-0.12.3.tar.gz (13 kB)
Collecting google-cloud-bigtable<2,>=0.31.1
  Downloading google_cloud_bigtable-1.6.1-py2.py3-none-any.whl (267
kB)
-v1<0.12dev,>=0.11.4
  Downloading grpc-google-iam-v1-0.11.4.tar.gz (12 kB)
Collecting google-cloud-bigtable<2,>=0.31.1
  Downloading google_cloud_bigtable-0.32.2-py2.py3-none-any.whl (156
kB)
ultiple versions of google-cloud-bigquery-storage to determine which
version is compatible with other requirements. This could take a
while.
Collecting google-cloud-bigquery-storage>=2.6.3
  Downloading google_cloud_bigquery_storage-2.12.0-py2.py3-none-
any.whl (179 kB)
-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.7 MB)
ultiple versions of google-cloud-bigquery-storage to determine which
version is compatible with other requirements. This could take a
while.
  Downloading google_cloud_bigquery_storage-2.7.0-py2.py3-none-any.whl
(125 kB)
ultiple versions of google-cloud-bigquery to determine which version
is compatible with other requirements. This could take a while.
Collecting google-cloud-bigquery<3,>=2.26.0
  Downloading google_cloud_bigquery-2.34.2-py2.py3-none-any.whl (206
kB)
ight need to provide the dependency resolver with stricter constraints
to reduce runtime. If you want to abort this run, you can press Ctrl +
C to do so. To improve how pip performs, tell us what happened here:
https://pip.pypa.io/surveys/backtracking
  Downloading google_cloud_bigquery-2.34.1-py2.py3-none-any.whl (206
kB)
ultiple versions of google-cloud-bigquery to determine which version
is compatible with other requirements. This could take a while.
  Downloading google_cloud_bigquery-2.30.0-py2.py3-none-any.whl (203
kB)
ight need to provide the dependency resolver with stricter constraints
to reduce runtime. If you want to abort this run, you can press Ctrl +
C to do so. To improve how pip performs, tell us what happened here:
https://pip.pypa.io/surveys/backtracking
  Downloading google_cloud_bigquery-2.27.0-py2.py3-none-any.whl (201
kB)
ultiple versions of google-api-core to determine which version is
compatible with other requirements. This could take a while.
```

```
INFO: pip is looking at multiple versions of google-api-core[grpc] to
determine which version is compatible with other requirements. This
could take a while.
Collecting google-api-core[grpc]!=2.0.*,!=2.1.*,!=2.2.*,!
=2.3.0,<3.0.0dev,>=1.31.5
  Downloading google_api_core-2.7.0-py3-none-any.whl (114 kB)
-manylinux_2_12_x86_64.manylinux2010_x86_64.whl (38 kB)
Collecting grpcio<2,>=1.28.1
  Downloading grpcio-1.45.0-cp37-cp37m-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl (4.4 MB)
ent already satisfied: docopt in /usr/local/lib/python3.7/dist-
packages (from hdfs<3.0.0,>=2.1.0->apache-beam[gcp]<3,>=2.36->tfx)
(0.6.2)
Requirement already satisfied: MarkupSafe>=0.23 in
/usr/local/lib/python3.7/dist-packages (from jinja2<4,>=2.7.3->tfx)
(2.0.1)
Requirement already satisfied: ipython in
/usr/local/lib/python3.7/dist-packages (from keras-tuner<2,>=1.0.4-
>tfx) (5.5.0)
Requirement already satisfied: tensorboard in
/usr/local/lib/python3.7/dist-packages (from keras-tuner<2,>=1.0.4-
>tfx) (2.8.0)
Collecting kt-legacy
  Downloading kt_legacy-1.0.4-py3-none-any.whl (9.6 kB)
Requirement already satisfied: requests-oauthlib in
/usr/local/lib/python3.7/dist-packages (from kubernetes<13,>=10.0.1-
>tfx) (1.3.1)
Requirement already satisfied: certifi>=14.05.14 in
/usr/local/lib/python3.7/dist-packages (from kubernetes<13,>=10.0.1-
>tfx) (2021.10.8)
Requirement already satisfied: urllib3>=1.24.2 in
/usr/local/lib/python3.7/dist-packages (from kubernetes<13,>=10.0.1-
>tfx) (1.24.3)
Requirement already satisfied: pyasn1>=0.1.7 in
/usr/local/lib/python3.7/dist-packages (from oauth2client<5,>=2.0.1-
>apache-beam[gcp]<3,>=2.36->tfx) (0.4.8)
Collecting typing-utils>=0.0.3
  Downloading typing_utils-0.1.0-py3-none-any.whl (10 kB)
Requirement already satisfied: pyparsing>=2.0.2 in
/usr/local/lib/python3.7/dist-packages (from packaging<21,>=20->tfx)
(3.0.7)
Collecting protobuf<4,>=3.13
  Downloading protobuf-3.20.0-cp37-cp37m-
manylinux_2_5_x86_64.manylinux1_x86_64.whl (1.0 MB)
ent already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.7/dist-
packages (from requests<3.0.0,>=2.24.0->apache-beam[gcp]<3,>=2.36-
>tfx) (2.10)
Requirement already satisfied: charset-normalizer~=2.0.0 in
/usr/local/lib/python3.7/dist-packages (from requests<3.0.0,>=2.24.0-
>apache-beam[gcp]<3,>=2.36->tfx) (2.0.12)
```

```
Requirement already satisfied: termcolor>=1.1.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
>tfx) (1.1.0)
Collecting tf-estimator-nightly==2.8.0.dev2021122109
  Downloading tf_estimator_nightly-2.8.0.dev2021122109-py2.py3-none-
any.whl (462 kB)
ent already satisfied: keras-preprocessing>=1.1.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
>tfx) (1.1.2)
Requirement already satisfied: tensorflow-io-gcs-filesystem>=0.23.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
>tfx) (0.24.0)
Requirement already satisfied: astunparse>=1.6.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
>tfx) (1.6.3)
Requirement already satisfied: wrapt>=1.11.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
>tfx) (1.14.0)
Requirement already satisfied: libclang>=9.0.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
>tfx) (13.0.0)
Requirement already satisfied: google-pasta>=0.1.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
>tfx) (0.2.0)
Requirement already satisfied: gast>=0.2.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
>tfx) (0.5.3)
Requirement already satisfied: keras<2.9,>=2.8.0rc0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
>tfx) (2.8.0)
Requirement already satisfied: opt-einsum>=2.3.2 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
>tfx) (3.3.0)
Requirement already satisfied: h5py>=2.9.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
>tfx) (3.1.0)
Requirement already satisfied: flatbuffers>=1.12 in
/usr/local/lib/python3.7/dist-packages (from tensorflow!=2.0.*,!
=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!=2.7.*,<2.9,>=1.15.5-
```

```
>tfx) (2.0)
Requirement already satisfied: wheel<1.0,>=0.23.0 in
/usr/local/lib/python3.7/dist-packages (from astunparse>=1.6.0-
>tensorflow!=2.0.*,!=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!
=2.7.*,<2.9,>=1.15.5->tfx) (0.37.1)
Requirement already satisfied: cached-property in
/usr/local/lib/python3.7/dist-packages (from h5py>=2.9.0->tensorflow!
=2.0.*,!=2.1.*,!=2.2.*,!=2.3.*,!=2.4.*,!=2.5.*,!=2.6.*,!
=2.7.*,<2.9,>=1.15.5->tfx) (1.5.2)
Requirement already satisfied: google-auth-oauthlib<0.5,>=0.4.1 in
/usr/local/lib/python3.7/dist-packages (from tensorboard->keras-
tuner<2,>=1.0.4->tfx) (0.4.6)
Requirement already satisfied: tensorboard-data-server<0.7.0,>=0.6.0
in /usr/local/lib/python3.7/dist-packages (from tensorboard->keras-
tuner<2,>=1.0.4->tfx) (0.6.1)
Requirement already satisfied: markdown>=2.6.8 in
/usr/local/lib/python3.7/dist-packages (from tensorboard->keras-
tuner<2,>=1.0.4->tfx) (3.3.6)
Requirement already satisfied: werkzeug>=0.11.15 in
/usr/local/lib/python3.7/dist-packages (from tensorboard->keras-
tuner<2,>=1.0.4->tfx) (1.0.1)
Requirement already satisfied: tensorboard-plugin-wit>=1.6.0 in
/usr/local/lib/python3.7/dist-packages (from tensorboard->keras-
tuner<2,>=1.0.4->tfx) (1.8.1)
Requirement already satisfied: importlib-metadata>=4.4 in
/usr/local/lib/python3.7/dist-packages (from markdown>=2.6.8-
>tensorboard->keras-tuner<2,>=1.0.4->tfx) (4.11.3)
Requirement already satisfied: zipp>=0.5 in
/usr/local/lib/python3.7/dist-packages (from importlib-metadata>=4.4-
>markdown>=2.6.8->tensorboard->keras-tuner<2,>=1.0.4->tfx) (3.7.0)
Requirement already satisfied: oauthlib>=3.0.0 in
/usr/local/lib/python3.7/dist-packages (from requests-oauthlib-
>kubernetes<13,>=10.0.1->tfx) (3.2.0)
Collecting joblib<0.15,>=0.12
  Downloading joblib-0.14.1-py2.py3-none-any.whl (294 kB)
hash<0.4,>=0.2
  Downloading pyfarmhash-0.3.2.tar.gz (99 kB)
ent already satisfied: pandas<2,>=1.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-data-
validation<1.8.0,>=1.7.0->tfx) (1.3.5)
Requirement already satisfied: tensorflow-metadata<1.8,>=1.7.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-data-
validation<1.8.0,>=1.7.0->tfx) (1.7.0)
Collecting ipython
  Downloading ipython-7.32.0-py3-none-any.whl (793 kB)
ent already satisfied: scipy<2,>=1.4.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-model-
analysis<0.39,>=0.38.0->tfx) (1.4.1)
Requirement already satisfied: ipywidgets<8,>=7 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-model-
```

```
analysis<0.39,>=0.38.0->tfx) (7.7.0)
Requirement already satisfied: pickleshare in
/usr/local/lib/python3.7/dist-packages (from ipython->keras-
tuner<2,>=1.0.4->tfx) (0.7.5)
Collecting prompt-toolkit!=3.0.0,!=3.0.1,<3.1.0,>=2.0.0
  Downloading prompt_toolkit-3.0.28-py3-none-any.whl (380 kB)
ent already satisfied: decorator in /usr/local/lib/python3.7/dist-
packages (from ipython->keras-tuner<2,>=1.0.4->tfx) (4.4.2)
Requirement already satisfied: pexpect>4.3 in
/usr/local/lib/python3.7/dist-packages (from ipython->keras-
tuner<2,>=1.0.4->tfx) (4.8.0)
Requirement already satisfied: pygments in
/usr/local/lib/python3.7/dist-packages (from ipython->keras-
tuner<2,>=1.0.4->tfx) (2.6.1)
Requirement already satisfied: matplotlib-inline in
/usr/local/lib/python3.7/dist-packages (from ipython->keras-
tuner<2,>=1.0.4->tfx) (0.1.3)
Requirement already satisfied: traitlets>=4.2 in
/usr/local/lib/python3.7/dist-packages (from ipython->keras-
tuner<2,>=1.0.4->tfx) (5.1.1)
Requirement already satisfied: jedi>=0.16 in
/usr/local/lib/python3.7/dist-packages (from ipython->keras-
tuner<2,>=1.0.4->tfx) (0.18.1)
Requirement already satisfied: backcall in
/usr/local/lib/python3.7/dist-packages (from ipython->keras-
tuner<2,>=1.0.4->tfx) (0.2.0)
Requirement already satisfied: jupyterlab-widgets>=1.0.0 in
/usr/local/lib/python3.7/dist-packages (from ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (1.1.0)
Requirement already satisfied: nbformat>=4.2.0 in
/usr/local/lib/python3.7/dist-packages (from ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (5.2.0)
Requirement already satisfied: ipython-genutils~=0.2.0 in
/usr/local/lib/python3.7/dist-packages (from ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (0.2.0)
Requirement already satisfied: ipykernel>=4.5.1 in
/usr/local/lib/python3.7/dist-packages (from ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (4.10.1)
Requirement already satisfied: widgetsnbextension~=3.6.0 in
/usr/local/lib/python3.7/dist-packages (from ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (3.6.0)
Requirement already satisfied: tornado>=4.0 in
/usr/local/lib/python3.7/dist-packages (from ipykernel>=4.5.1-
>ipywidgets<8,>=7->tensorflow-model-analysis<0.39,>=0.38.0->tfx)
(5.1.1)
Requirement already satisfied: jupyter-client in
/usr/local/lib/python3.7/dist-packages (from ipykernel>=4.5.1-
>ipywidgets<8,>=7->tensorflow-model-analysis<0.39,>=0.38.0->tfx)
(5.3.5)
Requirement already satisfied: parso<0.9.0,>=0.8.0 in
```

```
/usr/local/lib/python3.7/dist-packages (from jedi>=0.16->ipython-
>keras-tuner<2,>=1.0.4->tfx) (0.8.3)
Requirement already satisfied: jupyter-core in
/usr/local/lib/python3.7/dist-packages (from nbformat>=4.2.0-
>ipywidgets<8,>=7->tensorflow-model-analysis<0.39,>=0.38.0->tfx)
(4.9.2)
Requirement already satisfied: jsonschema!=2.5.0,>=2.4 in
/usr/local/lib/python3.7/dist-packages (from nbformat>=4.2.0-
>ipywidgets<8,>=7->tensorflow-model-analysis<0.39,>=0.38.0->tfx)
(4.3.3)
Requirement already satisfied: importlib-resources>=1.4.0 in
/usr/local/lib/python3.7/dist-packages (from jsonschema!=2.5.0,>=2.4-
>nbformat>=4.2.0->ipywidgets<8,>=7->tensorflow-model-
analysis<0.39,>=0.38.0->tfx) (5.4.0)
Requirement already satisfied: pyrsistent!=0.17.0,!=0.17.1,!
=0.17.2,>=0.14.0 in /usr/local/lib/python3.7/dist-packages (from
jsonschema!=2.5.0,>=2.4->nbformat>=4.2.0->ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (0.18.1)
Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.7/dist-packages (from pexpect>4.3->ipython-
>keras-tuner<2,>=1.0.4->tfx) (0.7.0)
Requirement already satisfied: wcwidth in
/usr/local/lib/python3.7/dist-packages (from prompt-toolkit!=3.0.0,!
=3.0.1,<3.1.0,>=2.0.0->ipython->keras-tuner<2,>=1.0.4->tfx) (0.2.5)
Requirement already satisfied: notebook>=4.4.1 in
/usr/local/lib/python3.7/dist-packages (from
widgetsnbextension~=3.6.0->ipywidgets<8,>=7->tensorflow-model-
analysis<0.39,>=0.38.0->tfx) (5.3.1)
Requirement already satisfied: Send2Trash in
/usr/local/lib/python3.7/dist-packages (from notebook>=4.4.1-
>widgetsnbextension~=3.6.0->ipywidgets<8,>=7->tensorflow-model-
analysis<0.39,>=0.38.0->tfx) (1.8.0)
Requirement already satisfied: nbconvert in
/usr/local/lib/python3.7/dist-packages (from notebook>=4.4.1-
>widgetsnbextension~=3.6.0->ipywidgets<8,>=7->tensorflow-model-
analysis<0.39,>=0.38.0->tfx) (5.6.1)
Requirement already satisfied: terminado>=0.8.1 in
/usr/local/lib/python3.7/dist-packages (from notebook>=4.4.1-
>widgetsnbextension~=3.6.0->ipywidgets<8,>=7->tensorflow-model-
analysis<0.39,>=0.38.0->tfx) (0.13.3)
Requirement already satisfied: pyzmq>=13 in
/usr/local/lib/python3.7/dist-packages (from jupyter-client-
>ipykernel>=4.5.1->ipywidgets<8,>=7->tensorflow-model-
analysis<0.39,>=0.38.0->tfx) (22.3.0)
Requirement already satisfied: bleach in
/usr/local/lib/python3.7/dist-packages (from nbconvert-
>notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (4.1.0)
Requirement already satisfied: defusedxml in
/usr/local/lib/python3.7/dist-packages (from nbconvert-
```

```
>notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (0.7.1)
Requirement already satisfied: mistune<2,>=0.8.1 in
/usr/local/lib/python3.7/dist-packages (from nbconvert-
>notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (0.8.4)
Requirement already satisfied: entrypoints>=0.2.2 in
/usr/local/lib/python3.7/dist-packages (from nbconvert-
>notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (0.4)
Requirement already satisfied: testpath in
/usr/local/lib/python3.7/dist-packages (from nbconvert-
>notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (0.6.0)
Requirement already satisfied: pandocfilters>=1.4.1 in
/usr/local/lib/python3.7/dist-packages (from nbconvert-
>notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (1.5.0)
Requirement already satisfied: webencodings in
/usr/local/lib/python3.7/dist-packages (from bleach->nbconvert-
>notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets<8,>=7-
>tensorflow-model-analysis<0.39,>=0.38.0->tfx) (0.5.1)
WARNING: The candidate selected for download or install is a yanked
version: 'grpcio' candidate (version 1.45.0 at
https://files.pythonhosted.org/packages/b8/ab/c7abc950e222c4cab5f7fd92
107f3b09553061f673f1a670e6569105f584/grpcio-1.45.0-cp37-cp37m-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl#sha256=cc135b77f384a84b
ac67a37947886986be136356446338d64160a30c85f20c6d (from
https://pypi.org/simple/grpcio/) (requires-python:>=3.6))
Reason for being yanked: Segfaults
Building wheels for collected packages: dill, google-apitools, grpc-
google-iam-v1, pyfarmhash
  Building wheel for dill (setup.py) ... e=dill-0.3.1.1-py3-none-
any.whl size=78544
sha256=6f3b2879d2a5b1009f9b0bb4ed81b1f5d2e85b2082b4c6e7b528f282f5caaa5
c
  Stored in directory:
/root/.cache/pip/wheels/a4/61/fd/c57e374e580aa78a45ed78d5859b3a44436af
17e22ca53284f
  Building wheel for google-apitools (setup.py) ... e=google_apitools-
0.5.31-py3-none-any.whl size=131039
sha256=2f61052628fcc70596ad54ac3e076a591547fbb84c7285d4dc0629550b71e8c
8
  Stored in directory:
/root/.cache/pip/wheels/19/b5/2f/1cc3cf2b31e7a9cd1508731212526d9550271
274d351c96f16
  Building wheel for grpc-google-iam-v1 (setup.py) ... -v1:
filename=grpc_google_iam_v1-0.12.3-py3-none-any.whl size=18515
sha256=5fff5917c63b4ae6951b2c911dc16bc4a6972ca61ac7cfbfac66bd3874d8efd
8
```

```
  Stored in directory:
/root/.cache/pip/wheels/b9/ee/67/2e444183030cb8d31ce8b34cee34a7afdbd3b
a5959ea846380
  Building wheel for pyfarmhash (setup.py) ... hash:
filename=pyfarmhash-0.3.2-cp37-cp37m-linux_x86_64.whl size=68407
sha256=489cfdf88f0632af72120d39a683588b5e5690a38f1329a664d3b6ce8f0686e
0
  Stored in directory:
/root/.cache/pip/wheels/53/58/7a/3b040f3a2ee31908e3be916e32660db6db536
21ce6eba838dc
Successfully built dill google-apitools grpc-google-iam-v1 pyfarmhash
Installing collected packages: protobuf, requests, pytz, packaging,
grpcio, attrs, prompt-toolkit, google-api-core, typing-utils, proto-
plus, ipython, grpcio-status, grpcio-gcp, grpc-google-iam-v1, google-
crc32c, tf-estimator-nightly, pymongo, pyarrow, overrides, orjson,
hdfs, google-resumable-media, google-cloud-pubsub, google-cloud-core,
fasteners, fastavro, dill, cloudpickle, google-cloud-vision, google-
cloud-videointelligence, google-cloud-spanner, google-cloud-
recommendations-ai, google-cloud-pubsublite, google-cloud-language,
google-cloud-dlp, google-cloud-bigtable, google-cloud-bigquery-
storage, google-cloud-bigquery, google-apitools, apache-beam,
websocket-client, tensorflow-serving-api, tfx-bsl, pyfarmhash, ml-
metadata, kt-legacy, joblib, google-cloud-storage, docker, tensorflow-
transform, tensorflow-model-analysis, tensorflow-data-validation, ml-
pipelines-sdk, kubernetes, keras-tuner, google-cloud-aiplatform, tfx
  Attempting uninstall: protobuf
    Found existing installation: protobuf 3.17.3
    Uninstalling protobuf-3.17.3:
      Successfully uninstalled protobuf-3.17.3
  Attempting uninstall: requests
    Found existing installation: requests 2.23.0
    Uninstalling requests-2.23.0:
      Successfully uninstalled requests-2.23.0
  Attempting uninstall: pytz
    Found existing installation: pytz 2018.9
    Uninstalling pytz-2018.9:
      Successfully uninstalled pytz-2018.9
  Attempting uninstall: packaging
    Found existing installation: packaging 21.3
    Uninstalling packaging-21.3:
      Successfully uninstalled packaging-21.3
  Attempting uninstall: grpcio
    Found existing installation: grpcio 1.44.0
    Uninstalling grpcio-1.44.0:
      Successfully uninstalled grpcio-1.44.0
  Attempting uninstall: attrs
    Found existing installation: attrs 21.4.0
    Uninstalling attrs-21.4.0:
      Successfully uninstalled attrs-21.4.0
  Attempting uninstall: prompt-toolkit
```

```
  Found existing installation: prompt-toolkit 1.0.18
  Uninstalling prompt-toolkit-1.0.18:
    Successfully uninstalled prompt-toolkit-1.0.18
Attempting uninstall: google-api-core
  Found existing installation: google-api-core 1.26.3
  Uninstalling google-api-core-1.26.3:
    Successfully uninstalled google-api-core-1.26.3
Attempting uninstall: ipython
  Found existing installation: ipython 5.5.0
  Uninstalling ipython-5.5.0:
    Successfully uninstalled ipython-5.5.0
Attempting uninstall: pymongo
  Found existing installation: pymongo 4.0.2
  Uninstalling pymongo-4.0.2:
    Successfully uninstalled pymongo-4.0.2
Attempting uninstall: pyarrow
  Found existing installation: pyarrow 6.0.1
  Uninstalling pyarrow-6.0.1:
    Successfully uninstalled pyarrow-6.0.1
Attempting uninstall: google-resumable-media
  Found existing installation: google-resumable-media 0.4.1
  Uninstalling google-resumable-media-0.4.1:
    Successfully uninstalled google-resumable-media-0.4.1
Attempting uninstall: google-cloud-core
  Found existing installation: google-cloud-core 1.0.3
  Uninstalling google-cloud-core-1.0.3:
    Successfully uninstalled google-cloud-core-1.0.3
Attempting uninstall: dill
  Found existing installation: dill 0.3.4
  Uninstalling dill-0.3.4:
    Successfully uninstalled dill-0.3.4
Attempting uninstall: cloudpickle
  Found existing installation: cloudpickle 1.3.0
  Uninstalling cloudpickle-1.3.0:
    Successfully uninstalled cloudpickle-1.3.0
Attempting uninstall: google-cloud-language
  Found existing installation: google-cloud-language 1.2.0
  Uninstalling google-cloud-language-1.2.0:
    Successfully uninstalled google-cloud-language-1.2.0
Attempting uninstall: google-cloud-bigquery-storage
  Found existing installation: google-cloud-bigquery-storage 1.1.0
  Uninstalling google-cloud-bigquery-storage-1.1.0:
    Successfully uninstalled google-cloud-bigquery-storage-1.1.0
Attempting uninstall: google-cloud-bigquery
  Found existing installation: google-cloud-bigquery 1.21.0
  Uninstalling google-cloud-bigquery-1.21.0:
    Successfully uninstalled google-cloud-bigquery-1.21.0
Attempting uninstall: joblib
  Found existing installation: joblib 1.1.0
  Uninstalling joblib-1.1.0:
```

```
      Successfully uninstalled joblib-1.1.0
  Attempting uninstall: google-cloud-storage
    Found existing installation: google-cloud-storage 1.18.1
    Uninstalling google-cloud-storage-1.18.1:
      Successfully uninstalled google-cloud-storage-1.18.1
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
pandas-gbq 0.13.3 requires google-cloud-
bigquery[bqstorage,pandas]<2.0.0dev,>=1.11.1, but you have google-
cloud-bigquery 2.34.3 which is incompatible.
multiprocess 0.70.12.2 requires dill>=0.3.4, but you have dill 0.3.1.1
which is incompatible.
jupyter-console 5.2.0 requires prompt-toolkit<2.0.0,>=1.0.0, but you
have prompt-toolkit 3.0.28 which is incompatible.
gym 0.17.3 requires cloudpickle<1.7.0,>=1.2.0, but you have
cloudpickle 2.0.0 which is incompatible.
google-colab 1.0.0 requires ipython~=5.5.0, but you have ipython
7.32.0 which is incompatible.
google-colab 1.0.0 requires requests~=2.23.0, but you have requests
2.27.1 which is incompatible.
datascience 0.10.6 requires folium==0.2.1, but you have folium 0.8.3
which is incompatible.
Successfully installed apache-beam-2.37.0 attrs-20.3.0 cloudpickle-
2.0.0 dill-0.3.1.1 docker-4.4.4 fastavro-1.4.10 fasteners-0.17.3
google-api-core-1.31.5 google-apitools-0.5.31 google-cloud-aiplatform-
1.11.0 google-cloud-bigquery-2.34.3 google-cloud-bigquery-storage-
2.13.0 google-cloud-bigtable-1.7.0 google-cloud-core-1.7.2 google-
cloud-dlp-3.6.2 google-cloud-language-1.3.0 google-cloud-pubsub-2.11.0
google-cloud-pubsublite-1.4.1 google-cloud-recommendations-ai-0.2.0
google-cloud-spanner-1.19.1 google-cloud-storage-2.2.1 google-cloud-
videointelligence-1.16.1 google-cloud-vision-1.0.0 google-crc32c-1.3.0
google-resumable-media-2.3.2 grpc-google-iam-v1-0.12.3 grpcio-1.45.0
grpcio-gcp-0.2.2 grpcio-status-1.45.0 hdfs-2.7.0 ipython-7.32.0
joblib-0.14.1 keras-tuner-1.1.2 kt-legacy-1.0.4 kubernetes-12.0.1 ml-
metadata-1.7.0 ml-pipelines-sdk-1.7.0 orjson-3.6.7 overrides-6.1.0
packaging-20.9 prompt-toolkit-3.0.28 proto-plus-1.20.3 protobuf-3.20.0
pyarrow-5.0.0 pyfarmhash-0.3.2 pymongo-3.12.3 pytz-2022.1 requests-
2.27.1 tensorflow-data-validation-1.7.0 tensorflow-model-analysis-
0.38.0 tensorflow-serving-api-2.8.0 tensorflow-transform-1.7.0 tf-
estimator-nightly-2.8.0.dev2021122109 tfx-1.7.0 tfx-bsl-1.7.0 typing-
utils-0.1.0 websocket-client-1.3.2
```


```
{"pip_warning":{"packages":["IPython","google","prompt_toolkit"]}}
```


```
from tfx import v1 as tfx
```

```
import tensorflow as tf
print('TensorFlow version: {}'.format(tf.__version__))
from tfx import v1 as tfx
print('TFX version: {}'.format(tfx.__version__))
```

```
TensorFlow version: 2.8.0
TFX version: 1.7.0
```

```
import os
path = '/content/drive/MyDrive/Colab Notebooks/week7'
PIPELINE_NAME = "penguin-simple"

# Output directory to store artifacts generated from the pipeline.
PIPELINE_ROOT = os.path.join(path,'pipelines', PIPELINE_NAME)
# Path to a SQLite DB file to use as an MLMD storage.
METADATA_PATH = os.path.join(path,'metadata', PIPELINE_NAME,
'metadata.db')
# Output directory where created models from the pipeline will be
exported.
SERVING_MODEL_DIR = os.path.join(path,'serving_model', PIPELINE_NAME)
print(PIPELINE_ROOT)
print(METADATA_PATH)
print(SERVING_MODEL_DIR)
from absl import logging
logging.set_verbosity(logging.INFO)  # Set default logging level.
```

```
/content/drive/MyDrive/Colab Notebooks/week7/pipelines/penguin-simple
/content/drive/MyDrive/Colab
Notebooks/week7/metadata/penguin-simple/metadata.db
/content/drive/MyDrive/Colab Notebooks/week7/serving_model/penguin-
simple
```

```
import urllib.request
import tempfile

DATA_ROOT = tempfile.mkdtemp(prefix='tfx-data')  # Create a temporary
directory.
_data_url =
'https://raw.githubusercontent.com/tensorflow/tfx/master/tfx/examples/
penguin/data/labelled/penguins_processed.csv'
_data_filepath = os.path.join(path,DATA_ROOT, "data.csv")
urllib.request.urlretrieve(_data_url, _data_filepath)
```

```
('/tmp/tfx-dataak3ax403/data.csv', <http.client.HTTPMessage at
0x7fb8d4074b50>)
```

```
print(_data_filepath)
```

```
/tmp/tfx-dataak3ax403/data.csv
```

```
!head {_data_filepath}
```

```
species,culmen_length_mm,culmen_depth_mm,flipper_length_mm,body_mass_g
0,0.2545454545454545,0.6666666666666666,0.15254237288135594,0.29166666
66666667
0,0.26909090909090905,0.5119047619047618,0.23728813559322035,0.3055555
555555556
```

```
0,0.29818181818181805,0.5833333333333334,0.3898305084745763,0.1527777777777778
0,0.16727272727272732,0.7380952380952381,0.3559322033898305,0.2083333333333334
0,0.26181818181818167,0.892857142857143,0.3050847457627119,0.2638888888888889
0,0.24727272727272717,0.5595238095238096,0.15254237288135594,0.2569444444444444
0,0.25818181818181823,0.773809523809524,0.3898305084745763,0.5486111111111112
0,0.32727272727272727,0.5357142857142859,0.1694915254237288,0.1388888888888889
0,0.23636363636363636,0.9642857142857142,0.3220338983050847,0.3055555555555556
```

Importing libraries

```
import tensorflow as tf
from tensorflow import keras

# Helper libraries
import numpy as np
import matplotlib.pyplot as plt
import os
import subprocess
```

Loading datasets

```
(train_images, train_labels), (test_images, test_labels) = tf.keras.datasets.cifar10.load_data()
```

```
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170500096/170498071 [==============================] - 4s 0us/step
170508288/170498071 [==============================] - 4s 0us/step
```

Normalize and one hot encoding

```
train_images = train_images / 255.0
test_images = test_images / 255.0
```

```
train_images.shape,test_images.shape
```

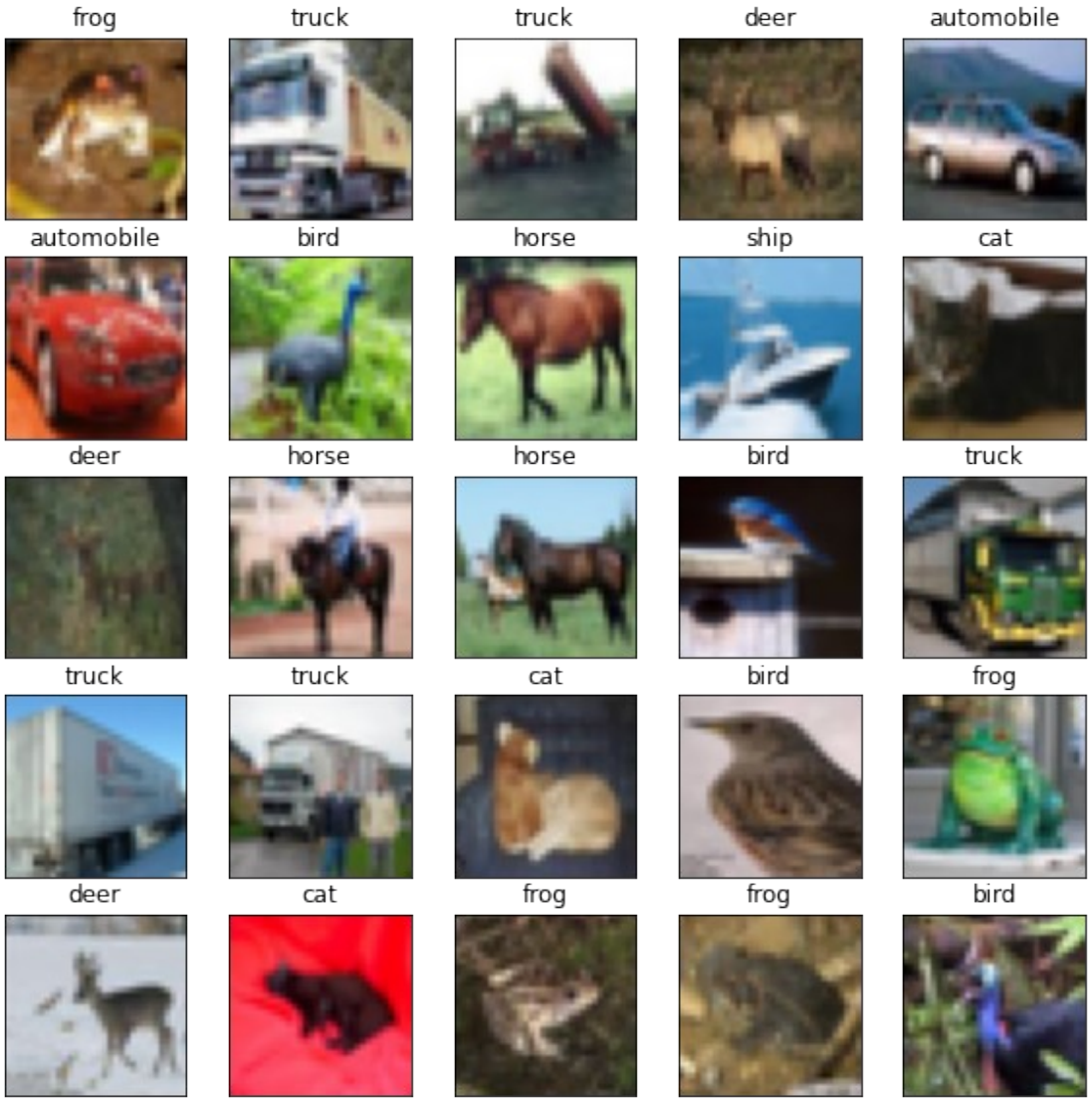
```
((50000, 32, 32, 3), (10000, 32, 32, 3))
```

```
import matplotlib.pyplot as plt

label_names = {
0 : 'airplane',
1 : 'automobile',
```

```
2 : 'bird',
3 : 'cat',
4 : 'deer',
5 : 'dog',
6 : 'frog',
7 : 'horse',
8 : 'ship',
9 : 'truck',
}

rows, columns = 5, 5
fig = plt.figure(figsize=(10,10))
for i in range(1,columns*rows+1):
  fig.add_subplot(rows, columns, i)
  plt.imshow(train_images[i-1])
  plt.xticks([])
  plt.yticks([])
  plt.title("{}".format(label_names[train_labels[i-1][0]]))
plt.show();
```




```
print('\ntrain_images.shape: {}, of {}'.format(train_images.shape,
train_images.dtype))
print('test_images.shape: {}, of {}'.format(test_images.shape,
test_images.dtype))

train_labels = tf.keras.utils.to_categorical(train_labels)
test_labels = tf.keras.utils.to_categorical(test_labels)
```

```
train_images.shape: (50000, 32, 32, 3), of float64
test_images.shape: (10000, 32, 32, 3), of float64
```

Callbacks

```
path = '/content/drive/MyDrive/Colab Notebooks/'
checkpoint = tf.keras.callbacks.ModelCheckpoint(path,monitor =
'val_loss', verbose = 1, save_best_only = True, mode = 'min')
```

```
stop_early = tf.keras.callbacks.EarlyStopping(monitor
='val_accuracy',mode = 'auto', patience = 5)
reduce_lr =
tf.keras.callbacks.ReduceLROnPlateau(monitor='val_loss',factor = 0.2,
patience = 5,mode = 'auto', min_lr = 0.0001)
callbacklist = [stop_early,reduce_lr,checkpoint]
```

Model

```
from tensorflow.keras.regularizers import l1,l2
weight_decay = 1e-4
batch_size = 64
initial_epochs = 10
lr = 0.001
lr_decay = 1e-6
def model_build():
  inputs = keras.Input(shape=(32, 32, 3))
  x = tf.keras.layers.Conv2D(filters = 32,
                             kernel_size = (3,3),
                             padding = 'same',
                             activation = 'relu',
                             kernel_regularizer = l2(weight_decay)
                             )(inputs)
  x = tf.keras.layers.BatchNormalization()(x)

  x = tf.keras.layers.Conv2D(filters = 32,
                             kernel_size = (3,3),
                             padding = 'same',
                             activation = 'relu',
                             kernel_regularizer = l2(weight_decay)
                             )(x)
  x = tf.keras.layers.BatchNormalization()(x)
  x = tf.keras.layers.MaxPool2D(pool_size=(2,2))(x)
  x = tf.keras.layers.Dropout(rate = 0.2)(x)

  x = tf.keras.layers.Conv2D(filters = 64,
                             kernel_size = (3,3),
                             padding = 'same',
                             activation = 'relu',
                             kernel_regularizer = l2(weight_decay)
                             )(x)
  x = tf.keras.layers.BatchNormalization()(x)

  x = tf.keras.layers.Conv2D(filters = 64,
                             kernel_size = (3,3),
                             padding = 'same',
                             activation = 'relu',
                             kernel_regularizer = l2(weight_decay)
                             )(x)
  x = tf.keras.layers.BatchNormalization()(x)
```

```
  x = tf.keras.layers.MaxPool2D(pool_size=(2,2))(x)

  x = tf.keras.layers.Dropout(rate = 0.3)(x)

  x = tf.keras.layers.Conv2D(filters = 128,
                             kernel_size = (3,3),
                             padding = 'same',
                             activation = 'relu',
                             kernel_regularizer =  l2(weight_decay)
                             )(x)
  x = tf.keras.layers.BatchNormalization()(x)

  x = tf.keras.layers.Conv2D(filters = 128,
                             kernel_size = (3,3),
                             padding = 'same',
                             activation = 'relu',
                             kernel_regularizer = l2(weight_decay)
                             )(x)
  x = tf.keras.layers.BatchNormalization()(x)

  x = tf.keras.layers.MaxPool2D(pool_size=(2,2))(x)

  x = tf.keras.layers.Dropout(rate = 0.4)(x)

  x = tf.keras.layers.Flatten()(x)

  x = tf.keras.layers.Dense(1024, activation = 'relu')(x)
  x = tf.keras.layers.Dropout(rate = 0.3)(x)

  x = tf.keras.layers.Dense(256, activation = 'relu')(x)
  x = tf.keras.layers.Dropout(rate = 0.2)(x)

  outputs = tf.keras.layers.Dense(10, activation = 'softmax')(x)
  model = tf.keras.Model(inputs = inputs,outputs=outputs, name
='cifar-10')

  model.compile(loss = tf.keras.losses.CategoricalCrossentropy(),
                optimizer = tf.keras.optimizers.Adam(learning_rate=
lr, decay = lr_decay),
                metrics=['accuracy'])
  return model

my_model = model_build()
my_model.summary()
```

```
Model: "cifar-10"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 input_1 (InputLayer)        [(None, 32, 32, 3)]       0
```

```
conv2d (Conv2D)              (None, 32, 32, 32)        896

batch_normalization (BatchN  (None, 32, 32, 32)       128
ormalization)

conv2d_1 (Conv2D)            (None, 32, 32, 32)        9248

batch_normalization_1 (Batc  (None, 32, 32, 32)       128
hNormalization)

max_pooling2d (MaxPooling2D  (None, 16, 16, 32)       0
)

dropout (Dropout)            (None, 16, 16, 32)        0

conv2d_2 (Conv2D)            (None, 16, 16, 64)        18496

batch_normalization_2 (Batc  (None, 16, 16, 64)       256
hNormalization)

conv2d_3 (Conv2D)            (None, 16, 16, 64)        36928

batch_normalization_3 (Batc  (None, 16, 16, 64)       256
hNormalization)

max_pooling2d_1 (MaxPooling  (None, 8, 8, 64)         0
2D)

dropout_1 (Dropout)          (None, 8, 8, 64)          0

conv2d_4 (Conv2D)            (None, 8, 8, 128)         73856

batch_normalization_4 (Batc  (None, 8, 8, 128)        512
hNormalization)

conv2d_5 (Conv2D)            (None, 8, 8, 128)         147584

batch_normalization_5 (Batc  (None, 8, 8, 128)        512
hNormalization)

max_pooling2d_2 (MaxPooling  (None, 4, 4, 128)        0
2D)

dropout_2 (Dropout)          (None, 4, 4, 128)         0

flatten (Flatten)            (None, 2048)              0

dense (Dense)                (None, 1024)              2098176
```

```
 dropout_3 (Dropout)         (None, 1024)              0

 dense_1 (Dense)             (None, 256)               262400

 dropout_4 (Dropout)         (None, 256)               0

 dense_2 (Dense)             (None, 10)                2570

=================================================================
Total params: 2,651,946
Trainable params: 2,651,050
Non-trainable params: 896
_________________________________________________________________
```

```
epochs = 10
batch_size = 64
h1 = my_model.fit(train_images,
                  train_labels,
                  validation_split=0.2,
                  epochs=epochs,

                  callbacks = callbacklist)
```

```
Epoch 1/10
1249/1250 [============================>.] - ETA: 0s - loss: 0.3517 -
accuracy: 0.9379
Epoch 1: val_loss did not improve from 0.63499
1250/1250 [==============================] - 37s 30ms/step - loss:
0.3518 - accuracy: 0.9378 - val_loss: 0.6360 - val_accuracy: 0.8652 -
lr: 2.0000e-04
Epoch 2/10
1249/1250 [============================>.] - ETA: 0s - loss: 0.3457 -
accuracy: 0.9381
Epoch 2: val_loss did not improve from 0.63499
1250/1250 [==============================] - 36s 29ms/step - loss:
0.3456 - accuracy: 0.9381 - val_loss: 0.6453 - val_accuracy: 0.8647 -
lr: 2.0000e-04
Epoch 3/10
1249/1250 [============================>.] - ETA: 0s - loss: 0.3376 -
accuracy: 0.9395
Epoch 3: val_loss did not improve from 0.63499
1250/1250 [==============================] - 35s 28ms/step - loss:
0.3375 - accuracy: 0.9395 - val_loss: 0.6426 - val_accuracy: 0.8642 -
lr: 2.0000e-04
Epoch 4/10
1248/1250 [============================>.] - ETA: 0s - loss: 0.3274 -
accuracy: 0.9426
Epoch 4: val_loss did not improve from 0.63499
1250/1250 [==============================] - 33s 26ms/step - loss:
0.3274 - accuracy: 0.9426 - val_loss: 0.6443 - val_accuracy: 0.8654 -
```

```
lr: 2.0000e-04
Epoch 5/10
1249/1250 [============================>.] - ETA: 0s - loss: 0.3216 -
accuracy: 0.9440
Epoch 5: val_loss improved from 0.63499 to 0.62526, saving model to
/content/drive/MyDrive/Colab Notebooks/
INFO:tensorflow:Assets written to: /content/drive/MyDrive/Colab
Notebooks/assets

INFO:tensorflow:Assets written to: /content/drive/MyDrive/Colab
Notebooks/assets

 1250/1250 [==============================] - 40s 32ms/step - loss:
0.3217 - accuracy: 0.9439 - val_loss: 0.6253 - val_accuracy: 0.8645 -
lr: 2.0000e-04
Epoch 6/10
1249/1250 [============================>.] - ETA: 0s - loss: 0.3115 -
accuracy: 0.9460
Epoch 6: val_loss did not improve from 0.62526
1250/1250 [==============================] - 39s 31ms/step - loss:
0.3115 - accuracy: 0.9459 - val_loss: 0.6446 - val_accuracy: 0.8653 -
lr: 2.0000e-04
Epoch 7/10
1249/1250 [============================>.] - ETA: 0s - loss: 0.3112 -
accuracy: 0.9458
Epoch 7: val_loss did not improve from 0.62526
1250/1250 [==============================] - 37s 30ms/step - loss:
0.3112 - accuracy: 0.9458 - val_loss: 0.6315 - val_accuracy: 0.8679 -
lr: 2.0000e-04
Epoch 8/10
1249/1250 [============================>.] - ETA: 0s - loss: 0.3023 -
accuracy: 0.9475
Epoch 8: val_loss did not improve from 0.62526
1250/1250 [==============================] - 37s 29ms/step - loss:
0.3023 - accuracy: 0.9475 - val_loss: 0.6407 - val_accuracy: 0.8611 -
lr: 2.0000e-04
Epoch 9/10
1249/1250 [============================>.] - ETA: 0s - loss: 0.2988 -
accuracy: 0.9489
Epoch 9: val_loss did not improve from 0.62526
1250/1250 [==============================] - 37s 30ms/step - loss:
0.2987 - accuracy: 0.9489 - val_loss: 0.6285 - val_accuracy: 0.8682 -
lr: 2.0000e-04
Epoch 10/10
1250/1250 [==============================] - ETA: 0s - loss: 0.2816 -
accuracy: 0.9537
Epoch 10: val_loss did not improve from 0.62526
1250/1250 [==============================] - 36s 29ms/step - loss:
0.2816 - accuracy: 0.9537 - val_loss: 0.6554 - val_accuracy: 0.8655 -
lr: 2.0000e-04
```

```
total_epochs = epochs + 30
h2 = my_model.fit(train_images,train_labels,
                    validation_split = 0.2,
                    initial_epoch = h1.epoch[-1],
                    epochs = total_epochs,
                    callbacks = callbacklist)
```

```
Epoch 10/40
1250/1250 [==============================] - ETA: 0s - loss: 0.2741 -
accuracy: 0.9560
Epoch 10: val_loss did not improve from 0.62526
1250/1250 [==============================] - 37s 30ms/step - loss:
0.2741 - accuracy: 0.9560 - val_loss: 0.6417 - val_accuracy: 0.8665 -
lr: 1.0000e-04
Epoch 11/40
1249/1250 [============================>.] - ETA: 0s - loss: 0.2701 -
accuracy: 0.9569
Epoch 11: val_loss did not improve from 0.62526
1250/1250 [==============================] - 36s 29ms/step - loss:
0.2701 - accuracy: 0.9569 - val_loss: 0.6311 - val_accuracy: 0.8666 -
lr: 1.0000e-04
Epoch 12/40
1250/1250 [==============================] - ETA: 0s - loss: 0.2644 -
accuracy: 0.9584
Epoch 12: val_loss did not improve from 0.62526
1250/1250 [==============================] - 36s 29ms/step - loss:
0.2644 - accuracy: 0.9584 - val_loss: 0.6401 - val_accuracy: 0.8698 -
lr: 1.0000e-04
Epoch 13/40
1250/1250 [==============================] - ETA: 0s - loss: 0.2588 -
accuracy: 0.9604
Epoch 13: val_loss did not improve from 0.62526
1250/1250 [==============================] - 36s 29ms/step - loss:
0.2588 - accuracy: 0.9604 - val_loss: 0.6404 - val_accuracy: 0.8683 -
lr: 1.0000e-04
Epoch 14/40
1249/1250 [============================>.] - ETA: 0s - loss: 0.2532 -
accuracy: 0.9616
Epoch 14: val_loss did not improve from 0.62526
1250/1250 [==============================] - 34s 27ms/step - loss:
0.2532 - accuracy: 0.9616 - val_loss: 0.6300 - val_accuracy: 0.8698 -
lr: 1.0000e-04
Epoch 15/40
1248/1250 [============================>.] - ETA: 0s - loss: 0.2510 -
accuracy: 0.9606
Epoch 15: val_loss did not improve from 0.62526
1250/1250 [==============================] - 34s 27ms/step - loss:
0.2509 - accuracy: 0.9607 - val_loss: 0.6383 - val_accuracy: 0.8687 -
lr: 1.0000e-04
Epoch 16/40
```

```
1250/1250 [==============================] - ETA: 0s - loss: 0.2488 -
accuracy: 0.9627
Epoch 16: val_loss did not improve from 0.62526
1250/1250 [==============================] - 33s 26ms/step - loss:
0.2488 - accuracy: 0.9627 - val_loss: 0.6410 - val_accuracy: 0.8700 -
lr: 1.0000e-04
Epoch 17/40
1249/1250 [============================>.] - ETA: 0s - loss: 0.2457 -
accuracy: 0.9625
Epoch 17: val_loss did not improve from 0.62526
1250/1250 [==============================] - 33s 26ms/step - loss:
0.2456 - accuracy: 0.9625 - val_loss: 0.6472 - val_accuracy: 0.8712 -
lr: 1.0000e-04
Epoch 18/40
1249/1250 [============================>.] - ETA: 0s - loss: 0.2410 -
accuracy: 0.9650
Epoch 18: val_loss did not improve from 0.62526
1250/1250 [==============================] - 32s 26ms/step - loss:
0.2410 - accuracy: 0.9650 - val_loss: 0.6412 - val_accuracy: 0.8694 -
lr: 1.0000e-04
Epoch 19/40
1249/1250 [============================>.] - ETA: 0s - loss: 0.2377 -
accuracy: 0.9641
Epoch 19: val_loss did not improve from 0.62526
1250/1250 [==============================] - 33s 26ms/step - loss:
0.2377 - accuracy: 0.9641 - val_loss: 0.6388 - val_accuracy: 0.8714 -
lr: 1.0000e-04
Epoch 20/40
1248/1250 [============================>.] - ETA: 0s - loss: 0.2385 -
accuracy: 0.9643
Epoch 20: val_loss did not improve from 0.62526
1250/1250 [==============================] - 32s 26ms/step - loss:
0.2385 - accuracy: 0.9642 - val_loss: 0.6361 - val_accuracy: 0.8696 -
lr: 1.0000e-04
Epoch 21/40
1248/1250 [============================>.] - ETA: 0s - loss: 0.2292 -
accuracy: 0.9675
Epoch 21: val_loss did not improve from 0.62526
1250/1250 [==============================] - 32s 26ms/step - loss:
0.2296 - accuracy: 0.9674 - val_loss: 0.6521 - val_accuracy: 0.8715 -
lr: 1.0000e-04
Epoch 22/40
1250/1250 [==============================] - ETA: 0s - loss: 0.2285 -
accuracy: 0.9660
Epoch 22: val_loss did not improve from 0.62526
1250/1250 [==============================] - 32s 26ms/step - loss:
0.2285 - accuracy: 0.9660 - val_loss: 0.6493 - val_accuracy: 0.8713 -
lr: 1.0000e-04
Epoch 23/40
1249/1250 [============================>.] - ETA: 0s - loss: 0.2275 -
```

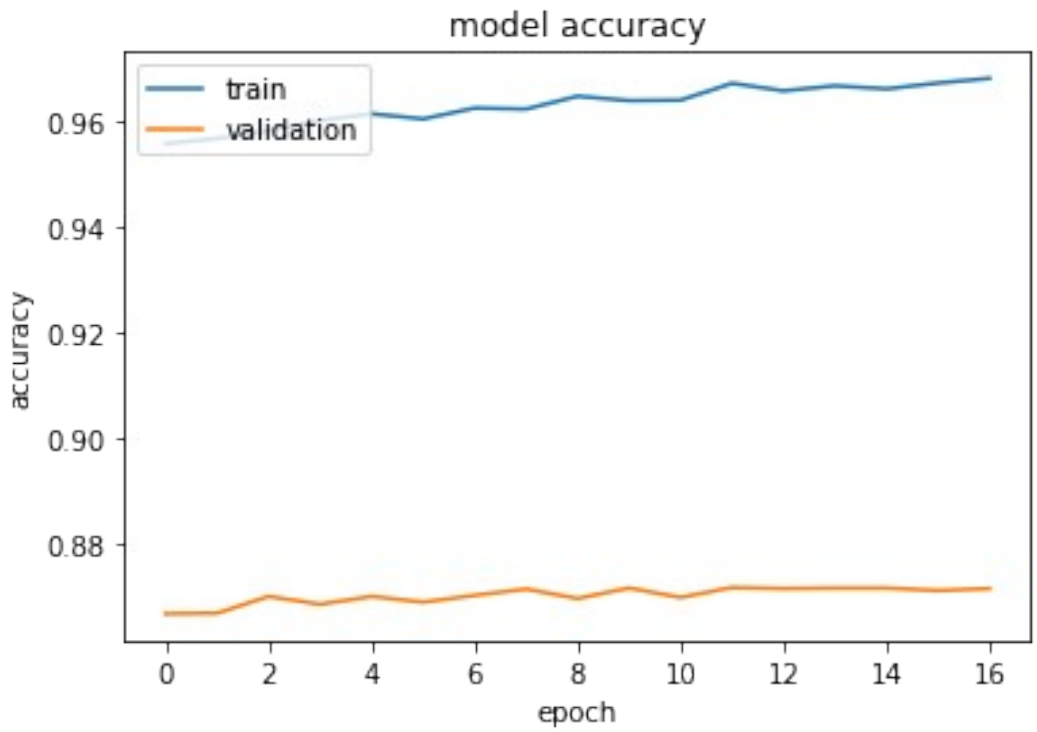
```
accuracy: 0.9669
Epoch 23: val_loss did not improve from 0.62526
1250/1250 [==============================] - 32s 25ms/step - loss:
0.2275 - accuracy: 0.9669 - val_loss: 0.6519 - val_accuracy: 0.8714 -
lr: 1.0000e-04
Epoch 24/40
1250/1250 [==============================] - ETA: 0s - loss: 0.2280 -
accuracy: 0.9664
Epoch 24: val_loss did not improve from 0.62526
1250/1250 [==============================] - 33s 26ms/step - loss:
0.2280 - accuracy: 0.9664 - val_loss: 0.6400 - val_accuracy: 0.8714 -
lr: 1.0000e-04
Epoch 25/40
1249/1250 [==============================>.] - ETA: 0s - loss: 0.2250 -
accuracy: 0.9675
Epoch 25: val_loss did not improve from 0.62526
1250/1250 [==============================] - 33s 26ms/step - loss:
0.2250 - accuracy: 0.9675 - val_loss: 0.6407 - val_accuracy: 0.8709 -
lr: 1.0000e-04
Epoch 26/40
1250/1250 [==============================] - ETA: 0s - loss: 0.2208 -
accuracy: 0.9684
Epoch 26: val_loss did not improve from 0.62526
1250/1250 [==============================] - 34s 27ms/step - loss:
0.2208 - accuracy: 0.9684 - val_loss: 0.6505 - val_accuracy: 0.8713 -
lr: 1.0000e-04
```

Performance

Accuracy

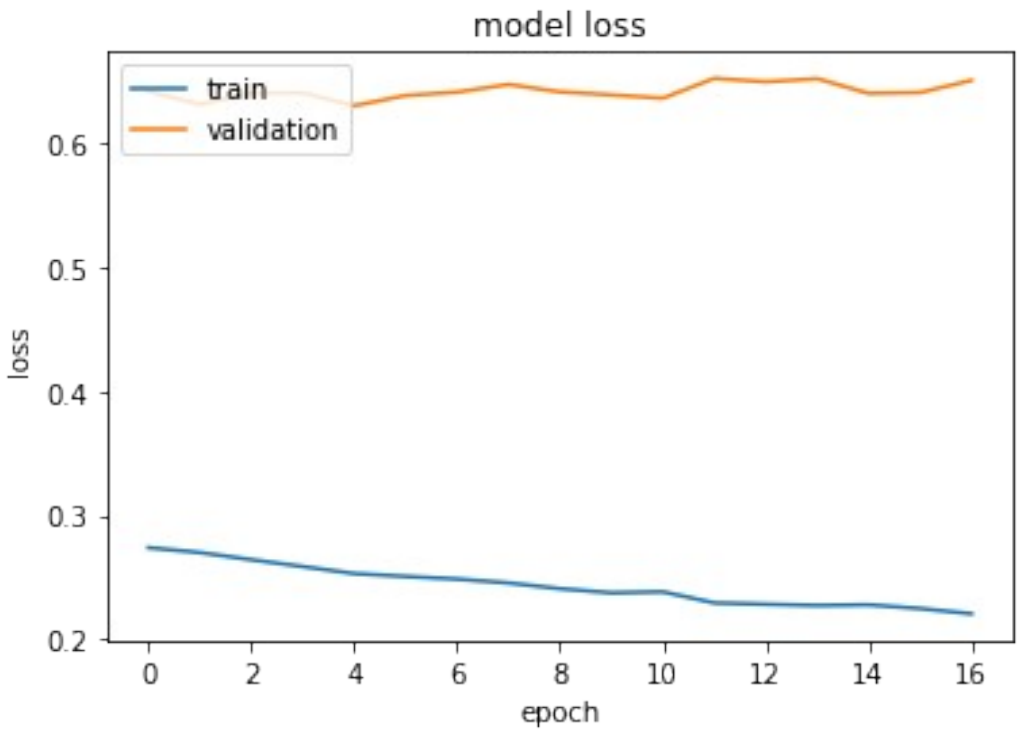
```
print(h2.history.keys)
plt.plot(h2.history['accuracy'])
plt.plot(h2.history['val_accuracy'])
plt.title('model accuracy')
plt.ylabel('accuracy')
plt.xlabel('epoch')
plt.legend(['train', 'validation'], loc='upper left')
plt.show()
```

```
<built-in method keys of dict object at 0x7fb7dcb2d0a0>
```



Loss

```
plt.plot(h2.history['loss'])
plt.plot(h2.history['val_loss'])
plt.title('model loss')
plt.ylabel('loss')
plt.xlabel('epoch')
plt.legend(['train', 'validation'], loc='upper left')
plt.show()
```



Model evaluation

```
test_loss, test_acc = my_model.evaluate(test_images, test_labels)
print('\nTest accuracy: {}'.format(test_acc))
```

```
313/313 [==============================] - 3s 9ms/step - loss: 0.6907
- accuracy: 0.8658

Test accuracy: 0.8658000230789185
```

Export path

```
import tensorflow

MODEL_DIR = tempfile.gettempdir()

version = 1
export_path = os.path.join(MODEL_DIR, str(version))

print('export_path = {}\n'.format(export_path))

tf.keras.models.save_model(
    my_model,
    export_path,
    overwrite = True,
    include_optimizer = True,
    save_format = None,
    signatures = None,
    options = None
```

```
)

print('\n Saved model:')
!ls-l {export_path}


export_path = /tmp/1

INFO:tensorflow:Assets written to: /tmp/1/assets

INFO:tensorflow:Assets written to: /tmp/1/assets


 Saved model:
/bin/bash: ls-l: command not found

!saved_model_cli show --dir {export_dir} --all

Traceback (most recent call last):
  File "/usr/local/bin/saved_model_cli", line 8, in <module>
    sys.exit(main())
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow/python/tools/saved_
model_cli.py", line 1260, in main
    args.func(args)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow/python/tools/saved_
model_cli.py", line 734, in show
    _show_all(args.dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow/python/tools/saved_
model_cli.py", line 302, in _show_all
    tag_sets =
saved_model_utils.get_saved_model_tag_sets(saved_model_dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow/python/tools/saved_
model_utils.py", line 87, in get_saved_model_tag_sets
    saved_model = read_saved_model(saved_model_dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow/python/tools/saved_
model_utils.py", line 51, in read_saved_model
    raise IOError("SavedModel file does not exist at: %s" %
saved_model_dir)
OSError: SavedModel file does not exist at: {export_dir}
```

Tensorflow model server

```
os.environ['MODEL_DIR'] = MODEL_DIR
```

TFServing

```
%%bash --bg
nohup tensorflow_model_server \
```

```
  --rest_api_port=8501 \
  --model_name=fashion_model \
  --model_base_path="${MODEL_DIR}" >server.log 2>&1

!tail server.log
```

Requests for TFServing

```
class_names = ['T-shirt/top', 'Trouser', 'Pullover', 'Dress', 'Coat',
               'Sandal', 'Shirt', 'Sneaker', 'Bag', 'Ankle boot']

def plot_img(ind,title):
  plt.figure()
  plt.imshow(test_images[ind])
  plt.axis('off')
  plt.title('\n\n{}'.format(title), fontdict={'size': 16})

import random
rando = random.randint(0,len(test_images)-1)
plot_img(rando, 'An Example Image:
{}'.format(label_names[train_labels[rando][0]]))
```



JSON

```
import json
data = json.dumps(
    {
        'signature_name':'serving_default',
        'instances':test_images[0:3].tolist()
```

```
    }
)

print('Data: {}'.format(data[:50],data[len(data)-52:]))
```

```
Data: {"signature_name": "serving_default", "instances":
```

REST request

```
!pip install -q requests
```

```
import requests
headers = {
    'content-type':'application/json'
}

json_response =
requests.post('http://localhost:8501/v1/models/fashion_model:predict',
data=data, headers=headers)
predictions = json.loads(json_response.text)['predictions']

plot_img(0, 'The model thought this was a {} (class {}), and it was
actually a {} (class {})'.format(
  class_names[np.argmax(predictions[0])], np.argmax(predictions[0]),
class_names[np.argmax(test_labels[0])], np.argmax(test_labels[0])))

np.argmax(predictions[0]),label_names[np.argmax(test_labels[0])]
```

```
(3, 'cat')
```



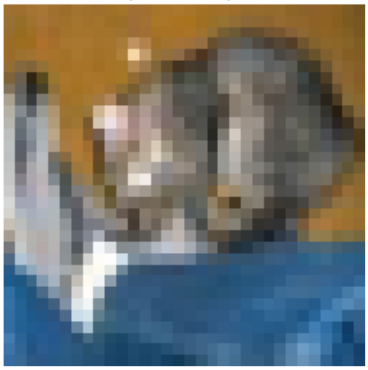
other versions

```
headers = {"content-type": "application/json"}
```

```
json_response =
requests.post('http://localhost:8501/v1/models/fashion_model/versions/
1:predict', data=data, headers=headers)
predictions = json.loads(json_response.text)['predictions']

for i in range(0,3):
  plot_img(i, 'The model thought this was a {} (class {}), and it was
actually a {} (class {})'.format(
  class_names[np.argmax(predictions[i])], np.argmax(predictions[i]),
class_names[np.argmax(test_labels[i])], np.argmax(test_labels[i])))
```

The model thought this was a Dress (class 3), and it was actually a Dress (class 3)



The model thought this was a Bag (class 8), and it was actually a Bag (class 8)



The model thought this was a Bag (class 8), and it was actually a Bag (class 8)

